BIO5101 (Introduction To Biotechnology) Complete Note (Topic 021-030)

BIO5101 – Introduction to Biotechnology

Topic 021 – Ligases and DNA Modification Enzymes

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1. Introduction

DNA ligases and **DNA modification enzymes** are essential tools in molecular biology used for **joining DNA fragments and modifying DNA molecules**. They play a crucial role in **recombinant DNA technology, cloning, and genetic engineering**.

Key Idea:

- DNA ligases **seal nicks in DNA**, forming phosphodiester bonds between adjacent nucleotides.
- DNA modification enzymes **alter DNA structure**, including methylation, phosphorylation, and labeling.

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2. DNA Ligases

1. **Function**
   - Catalyze formation of **phosphodiester bonds** between 3'-OH and 5'-phosphate ends of DNA.
   - Essential for **joining sticky or blunt ends** in cloning.
2. **Types of DNA Ligases**
   - **T4 DNA Ligase**: Most commonly used in labs; ligates both sticky and blunt ends.
   - **E. coli DNA Ligase**: Uses NAD^+; mainly ligates nicks in DNA.
   - **Thermostable DNA Ligase**: Used in **PCR-based ligation reactions**.
3. **Applications**
   - Joining DNA fragments in recombinant DNA technology.
   - Ligation of plasmid vectors and inserts.
   - Repair of DNA nicks during replication in vitro.

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3. DNA Modification Enzymes

1. **DNA Methyltransferases (MTases)**
   - Add methyl groups to specific DNA bases (usually adenine or cytosine).
   - Protect bacterial DNA from restriction enzymes.
   - Important for **epigenetic studies**.
2. **Kinases**
   - Add phosphate groups to 5'-ends of DNA or RNA.
   - Important for preparing DNA for ligation.
3. **Phosphatases**
   - Remove phosphate groups from DNA ends.
   - Prevent unwanted ligation in cloning experiments.
4. **Terminal Deoxynucleotidyl Transferase (TdT)**
   - Adds nucleotides to the 3'-ends of DNA without a template.
   - Used in labeling DNA and creating homopolymeric tails.

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4. Mechanism of DNA Ligation

1. **Activation** – DNA ligase forms an enzyme-adenylate intermediate.
2. **Binding** – Ligase binds to DNA fragments with compatible ends.
3. **Phosphodiester Bond Formation** – Ligase joins 3'-OH and 5'-phosphate ends.
4. **Release** – Newly ligated DNA fragment is released.

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5. Applications of Ligases and DNA Modification Enzymes

1. **Recombinant DNA Technology**
   - Joining plasmid vectors with inserts.
2. **Gene Cloning**
   - Construction of recombinant plasmids.
3. **DNA Repair Studies**
   - Studying in vitro repair mechanisms.
4. **Epigenetic Research**
   - Methyltransferases for studying gene regulation.
5. **DNA Labeling and Probing**
   - TdT and kinases for labeling DNA fragments in molecular diagnostics.

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6. Advantages

- Precise joining of DNA fragments.
- Enables manipulation of DNA for cloning, expression, and diagnostics.
- Compatible with various DNA ends (sticky and blunt).

7. Limitations

- Blunt-end ligation is less efficient than sticky-end ligation.
- Requires compatible ends for efficient ligation.
- Some enzymes are sensitive to temperature and reaction conditions.
- In vitro ligation may be affected by inhibitors or contaminants.

8. Summary Table

Enzyme	Function	Application	Example
DNA Ligase	Joins DNA fragments	Cloning, recombinant DNA	T4 DNA Ligase
DNA Methyltransferase	Adds methyl groups	Protect DNA, epigenetics	Dam MTase
Kinase	Adds phosphate groups	Prepare DNA for ligation	T4 Polynucleotide Kinase
Phosphatase	Removes phosphate groups	Prevent unwanted ligation	Calf Intestinal Phosphatase
TdT	Adds nucleotides to 3'-end	DNA labeling	Terminal deoxynucleotidyl transferase

Assessment Questions for Topic 021

A. Multiple Choice Questions (24 × 1 mark each)

1. DNA ligases catalyze:
   a) DNA cleavage
   b) Phosphodiester bond formation ✅
   c) DNA methylation
   d) DNA unwinding
2. T4 DNA ligase is commonly used because:
   a) Only ligates nicks
   b) Can ligate sticky and blunt ends ✅
   c) Only works at high temperature
   d) Cleaves DNA

3. DNA methyltransferases add:
   a) Phosphate groups
   b) Methyl groups ✅
   c) Hydroxyl groups
   d) Amino groups
4. Kinases function to:
   a) Remove phosphate groups
   b) Add phosphate groups ✅
   c) Ligate DNA
   d) Methylate DNA
5. Phosphatases are used to:
   a) Add methyl groups
   b) Remove phosphate groups ✅
   c) Ligate DNA
   d) Label DNA
6. TdT adds nucleotides to:
   a) 5'-end
   b) 3'-end ✅
   c) Internal positions
   d) RNA only
7. DNA ligase is essential for:
   a) DNA cleavage
   b) Joining DNA fragments ✅
   c) DNA replication only
   d) DNA transcription
8. Recombinant DNA construction requires:
   a) Ligase and restriction enzyme ✅
   b) Only ligase
   c) Only polymerase
   d) Only ribosomes
9. Blunt-end ligation is:
   a) More efficient than sticky ends
   b) Less efficient than sticky ends ✅
   c) Impossible
   d) Only occurs in RNA
10. Sticky ends are produced by:
    a) DNA methylation
    b) Restriction enzymes ✅
    c) Ligases
    d) Kinases
11. DNA ligase requires:
    a) ATP or NAD^+ ✅
    b) Only water
    c) Only buffer
    d) Only Mg^{2+}

12. Epigenetic studies use:
    a) TdT
    b) Methyltransferases ✅
    c) Ligase only
    d) Restriction enzymes only
13. DNA ligase mechanism involves:
    a) Hydrolysis
    b) Formation of enzyme-DNA adenylate ✅
    c) DNA methylation
    d) RNA ligation
14. DNA ligase joins:
    a) Protein fragments
    b) DNA fragments ✅
    c) RNA fragments
    d) Lipid fragments
15. Kinases are important for:
    a) Phosphorylation of DNA ends ✅
    b) Ligation
    c) Methylation
    d) Protein folding
16. Phosphatases prevent:
    a) DNA cleavage
    b) Unwanted ligation ✅
    c) DNA replication
    d) RNA synthesis
17. TdT is used in:
    a) DNA labeling ✅
    b) RNA transcription
    c) Protein degradation
    d) Lipid synthesis
18. DNA modification enzymes include:
    a) Ligases
    b) Methyltransferases, kinases, phosphatases, TdT ✅
    c) Ribosomes only
    d) Polymerases only
19. T4 DNA ligase uses:
    a) NAD^+
    b) ATP ✅
    c) Mg^{2+} only
    d) Water
20. Ligation efficiency is higher for:
    a) Sticky ends ✅
    b) Blunt ends
    c) RNA fragments
    d) Proteins

21. Recombinant plasmid formation requires:
    a) Ligase and restriction enzyme ✅
    b) Only ligase
    c) Only kinase
    d) Only phosphatase
22. DNA methylation can:
    a) Block restriction enzyme activity ✅
    b) Enhance ligation
    c) Remove phosphate groups
    d) Add nucleotides
23. Limitations of ligases include:
    a) Sensitive to reaction conditions ✅
    b) Can't join sticky ends
    c) Only works with RNA
    d) Cannot seal nicks
24. Applications of DNA modification enzymes include:
    a) Cloning, labeling, epigenetics ✅
    b) Protein folding only
    c) Lipid metabolism only
    d) RNA splicing only

B. Short Questions (6 × 3 marks each)

1. Define DNA ligase and explain its function.
2. Differentiate between sticky-end and blunt-end ligation.
3. Explain the role of DNA methyltransferases in molecular biology.
4. Describe the function of kinases and phosphatases in DNA modification.
5. What is terminal deoxynucleotidyl transferase and its application?
6. List three applications of DNA ligases and DNA modification enzymes.

C. Long Questions (6 × 5 marks each)

1. Explain the mechanism of DNA ligation and the role of ligases.
2. Describe the types of DNA ligases and their applications in biotechnology.
3. Discuss DNA modification enzymes and their functions in molecular biology.
4. Explain the difference between sticky-end and blunt-end ligation with examples.
5. Describe applications of DNA ligases and DNA modification enzymes in cloning and recombinant DNA technology.
6. Discuss limitations of ligases and DNA modification enzymes and strategies to overcome them.

BIO5101 – Introduction to Biotechnology

Topic 022 – Polymerases and Their Applications

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1. Introduction

Polymerases are enzymes that **synthesize nucleic acids (DNA or RNA) from nucleotides**. They are **essential tools in molecular biology and biotechnology** for DNA replication, amplification, and transcription.

Key Idea:

- DNA polymerases synthesize DNA.
- RNA polymerases synthesize RNA.
- Polymerases are used in **PCR, cloning, sequencing, and genetic engineering**.

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2. Types of Polymerases

A. DNA Polymerases

1. **Prokaryotic DNA Polymerases**
    - **DNA Pol I**: Removes RNA primers, fills gaps in lagging strand.
    - **DNA Pol II**: DNA repair polymerase.
    - **DNA Pol III**: Main enzyme for DNA replication.
2. **Eukaryotic DNA Polymerases**
    - **Pol α (alpha)**: Initiates DNA synthesis with primase.
    - **Pol δ (delta)**: Lagging strand synthesis.
    - **Pol ε (epsilon)**: Leading strand synthesis.
3. **Special DNA Polymerases**
    - **Taq DNA Polymerase**: Thermostable, used in PCR.
    - **Pfu Polymerase**: High-fidelity, proofreading enzyme.
    - **Reverse Transcriptase**: Synthesizes DNA from RNA template.

B. RNA Polymerases

1. **Prokaryotic RNA Polymerase**
    - Synthesizes mRNA, rRNA, and tRNA in bacteria.
2. **Eukaryotic RNA Polymerases**
    - **RNA Pol I**: rRNA synthesis.
    - **RNA Pol II**: mRNA synthesis.
    - **RNA Pol III**: tRNA and small RNA synthesis.

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3. Functions of Polymerases

- **DNA Replication**: Copying genomic DNA in vivo and in vitro.
- **DNA Amplification**: PCR using DNA polymerases.
- **Reverse Transcription**: cDNA synthesis from RNA templates.
- **DNA Repair**: Filling gaps and proofreading.
- **Transcription**: RNA polymerases synthesize RNA from DNA.

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4. Mechanism of DNA Polymerases

1. **Template Binding** – Polymerase binds to single-stranded DNA template.
2. **Primer Requirement** – DNA synthesis requires a primer with a free 3'-OH group.
3. **Nucleotide Incorporation** – Adds nucleotides complementary to the template strand.
4. **Proofreading (for some polymerases)** – 3'→5' exonuclease activity corrects errors.
5. **Extension** – Continues elongation until template is fully copied.

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5. Applications of Polymerases

1. **Polymerase Chain Reaction (PCR)**
   - Amplification of DNA for cloning, diagnostics, and research.
2. **cDNA Synthesis**
   - Reverse transcriptase synthesizes DNA from mRNA templates.
3. **DNA Sequencing**
   - DNA polymerases incorporate nucleotides for Sanger sequencing.
4. **Cloning and Genetic Engineering**
   - Filling in DNA gaps, ligation-ready fragments.
5. **Mutation Studies**
   - Polymerases with error-prone or high-fidelity activity for site-directed mutagenesis.
6. **Diagnostics**
   - PCR-based detection of pathogens, genetic disorders.

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6. Advantages

- High specificity and accuracy (especially proofreading polymerases).
- Thermostable polymerases allow PCR at high temperatures.
- Widely used in molecular biology and clinical diagnostics.

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7. Limitations

- Some polymerases lack proofreading (e.g., Taq polymerase).
- Require primers and templates.
- Contamination can lead to false results in PCR.
- Certain templates (high GC content) are difficult to amplify.

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8. Summary Table

Polymerase	Function	Application	Example
DNA Pol I	Primer removal, DNA repair	DNA replication	E. coli DNA Pol I
DNA Pol III	Main replication	Chromosomal replication	E. coli DNA Pol III
Taq DNA Pol	Thermostable PCR	PCR	Thermus aquaticus DNA Pol
Pfu Pol	High-fidelity PCR	Cloning, mutagenesis	Pyrococcus furiosus
Reverse Transcriptase	RNA → cDNA	cDNA synthesis	MMLV RT
RNA Pol II	mRNA synthesis	Gene expression studies	Eukaryotic RNA Pol II

Assessment Questions for Topic 022

A. Multiple Choice Questions (24 × 1 mark each)

1. DNA polymerases synthesize:
   a) RNA
   b) DNA ✅
   c) Proteins
   d) Lipids
2. RNA polymerases synthesize:
   a) DNA
   b) RNA ✅
   c) Proteins
   d) Lipids
3. Taq DNA polymerase is used in:
   a) DNA repair

b) PCR ✅
c) mRNA synthesis
d) Protein translation

4. Reverse transcriptase synthesizes:
   a) RNA from DNA
   b) DNA from RNA ✅
   c) RNA from RNA
   d) Protein from DNA
5. DNA polymerase requires:
   a) Template and primer ✅
   b) Only template
   c) Only primer
   d) Only ATP
6. DNA Pol III is:
   a) Main enzyme for replication ✅
   b) DNA repair only
   c) RNA synthesis
   d) Cloning only
7. DNA Pol I removes:
   a) Primers ✅
   b) RNA only
   c) DNA only
   d) Proteins
8. Proofreading activity of polymerases is:
   a) 5'→3'
   b) 3'→5' exonuclease ✅
   c) Ligase activity
   d) Methylation
9. RNA Pol II synthesizes:
   a) tRNA
   b) mRNA ✅
   c) rRNA
   d) miRNA
10. DNA polymerases are used in:
    a) PCR ✅
    b) Protein folding
    c) Lipid metabolism
    d) RNA splicing
11. High-fidelity polymerases are used in:
    a) Random mutation
    b) Accurate cloning ✅
    c) RNA synthesis
    d) Protein translation
12. Reverse transcriptase is derived from:
    a) Bacteria

b) Retroviruses ✅
c) Humans only
d) Plants

13. PCR requires:
    a) DNA polymerase ✅
    b) RNA polymerase
    c) Ligase
    d) Phosphatase
14. Taq polymerase is thermostable because:
    a) It comes from thermophilic bacteria ✅
    b) It is engineered in lab
    c) All polymerases are thermostable
    d) Only used at room temp
15. DNA polymerases extend DNA in:
    a) 3'→5'
    b) 5'→3' ✅
    c) Both directions
    d) Randomly
16. cDNA synthesis is used for:
    a) Protein synthesis
    b) RNA → DNA conversion ✅
    c) DNA degradation
    d) Lipid metabolism
17. Pfu polymerase is preferred when:
    a) High-fidelity is required ✅
    b) Low accuracy is acceptable
    c) RNA synthesis only
    d) Protein synthesis only
18. PCR amplification is limited by:
    a) Contamination ✅
    b) Only template sequence
    c) Only enzyme type
    d) All polymerases
19. Polymerases play a role in:
    a) DNA replication ✅
    b) Protein folding
    c) Lipid metabolism
    d) RNA splicing
20. DNA polymerases require:
    a) Nucleotides ✅
    b) Amino acids
    c) Lipids
    d) Proteins
21. RNA polymerase does not require:
    a) Template

b) Primer ✅
c) Nucleotides
d) Mg^{2+}

22. PCR is used in:
    a) Genetic testing ✅
    b) Lipid synthesis
    c) Protein translation
    d) Photosynthesis
23. High GC content in DNA may:
    a) Inhibit PCR ✅
    b) Facilitate PCR
    c) Enhance transcription
    d) Prevent translation
24. DNA polymerases are essential for:
    a) Recombinant DNA, PCR, sequencing ✅
    b) Protein folding only
    c) Lipid metabolism only
    d) RNA splicing only

B. Short Questions (6 × 3 marks each)

1. Define DNA and RNA polymerases and their functions.
2. Differentiate between prokaryotic and eukaryotic DNA polymerases.
3. Explain the role of Taq DNA polymerase in PCR.
4. Describe reverse transcriptase and its applications.
5. List three applications of polymerases in biotechnology.
6. Explain proofreading activity in DNA polymerases.

C. Long Questions (6 × 5 marks each)

1. Explain the types and functions of DNA polymerases.
2. Describe RNA polymerases and their role in transcription.
3. Discuss the mechanism of DNA synthesis by DNA polymerases.
4. Explain the applications of polymerases in PCR, cloning, and sequencing.
5. Discuss reverse transcriptase and its applications in cDNA synthesis and molecular diagnostics.
6. Explain limitations of polymerases in biotechnology and strategies to overcome them.

BIO5101 – Introduction to Biotechnology

Topic 023 – Gel Electrophoresis and DNA Separation Techniques

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1. Introduction

Gel electrophoresis is a **laboratory technique used to separate DNA, RNA, or proteins based on size and charge**. It is a **fundamental method in molecular biology and biotechnology** for analyzing nucleic acids, checking PCR products, and preparing DNA for cloning or sequencing.

Key Idea:

- Molecules move in an **electric field** through a gel matrix.
- Smaller fragments migrate faster, larger fragments migrate slower.

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2. Principles of Gel Electrophoresis

1. **Charge-Based Separation**
   - DNA and RNA are negatively charged due to **phosphate groups**.
   - They move towards the **positive electrode (anode)** in an electric field.
2. **Size-Based Separation**
   - Gel acts as a **molecular sieve**.
   - Smaller fragments move faster; larger fragments move slower.
3. **Gel Matrix**
   - **Agarose Gel**: Used for DNA and RNA (0.7–2% w/v).
   - **Polyacrylamide Gel (PAGE)**: Used for small DNA fragments, RNA, or proteins.
4. **Visualization**
   - DNA dyes: **Ethidium bromide (EtBr)**, **SYBR Green**.
   - Fluorescence under UV light allows fragment detection.

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3. Types of Gel Electrophoresis

1. **Agarose Gel Electrophoresis**
   - Separates **DNA fragments from ~100 bp to >20 kb**.
   - Easy to prepare and run.
2. **Polyacrylamide Gel Electrophoresis (PAGE)**
   - High resolution for **small DNA fragments (1–500 bp) or proteins**.
   - Can be denaturing (urea) or non-denaturing.
3. **Pulsed-Field Gel Electrophoresis (PFGE)**
   - Separates **very large DNA fragments (>50 kb)**.
   - Used in genomic mapping and bacterial typing.

4. **Capillary Electrophoresis**
    - Automated and high-throughput method.
    - Used in DNA sequencing and forensic analysis.

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4. Procedure of Agarose Gel Electrophoresis

1. **Gel Preparation**
    - Dissolve agarose in buffer, heat, and pour into a casting tray.
    - Insert comb to create wells.
2. **Sample Preparation**
    - Mix DNA with **loading dye** for visibility and density.
3. **Loading and Running**
    - Load DNA into wells.
    - Apply electric current; DNA migrates toward anode.
4. **Visualization**
    - Stain gel with EtBr or use pre-stained gels.
    - Visualize DNA bands under UV light.

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5. Factors Affecting Separation

- **Gel Concentration** – Higher % gel better for small fragments.
- **Voltage** – Higher voltage increases migration but can cause smearing.
- **Buffer** – Maintains pH and conductivity (e.g., TAE, TBE).
- **DNA Size** – Larger DNA moves slower.
- **DNA Conformation** – Supercoiled DNA migrates differently than linear DNA.

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6. Applications of Gel Electrophoresis

1. **DNA Analysis**
    - Checking PCR products, restriction digests, and plasmid preparations.
2. **RNA Analysis**
    - Assessing RNA integrity for RT-PCR or Northern blot.
3. **Protein Analysis**
    - SDS-PAGE separates proteins based on size.
4. **Genetic Fingerprinting**
    - STR analysis in forensics.
5. **Molecular Cloning**
    - Purifying DNA fragments for ligation.
6. **Diagnostics**
    - Detecting mutations, pathogens, or genetic disorders.

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7. Advantages

- Simple and cost-effective.
- High resolution for small to medium DNA fragments.
- Visualizes multiple samples simultaneously.
- Compatible with downstream applications.

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8. Limitations

- Cannot separate very large DNA fragments efficiently without PFGE.
- Requires DNA staining for visualization.
- Ethidium bromide is toxic and mutagenic.
- Time-consuming for high-resolution separation of many samples.

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9. Summary Table

Parameter	Description	Notes
Gel Type	Agarose / Polyacrylamide	Agarose for DNA, PAGE for small fragments/proteins
Separation Basis	Size and charge	Smaller fragments move faster
Visualization	EtBr, SYBR Green	UV light detection
Voltage	3–10 V/cm typical	Higher voltage increases speed but may cause smearing
Buffer	TAE / TBE	Maintains pH and conductivity
Application	DNA, RNA, proteins	PCR verification, cloning, diagnostics

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Assessment Questions for Topic 023

A. Multiple Choice Questions (24 × 1 mark each)

1. Gel electrophoresis separates molecules based on:
   a) Only charge
   b) Only size
   c) Charge and size ✅
   d) Only density

2. DNA migrates toward:
   a) Negative electrode
   b) Positive electrode ✅
   c) Any direction
   d) Center of gel
3. Agarose gel is typically used for:
   a) RNA only
   b) DNA ✅
   c) Proteins only
   d) Lipids
4. Polyacrylamide gel (PAGE) is used for:
   a) Large DNA fragments
   b) Small DNA fragments and proteins ✅
   c) Lipids
   d) Carbohydrates
5. PFGE is used to separate:
   a) Small DNA fragments
   b) Very large DNA fragments ✅
   c) RNA only
   d) Proteins only
6. DNA is negatively charged due to:
   a) Deoxyribose
   b) Phosphate groups ✅
   c) Nitrogenous bases
   d) Sugar backbone
7. Loading dye is used to:
   a) Visualize DNA ✅
   b) Digest DNA
   c) Inhibit migration
   d) Stain proteins
8. Ethidium bromide binds to:
   a) Proteins
   b) DNA ✅
   c) Lipids
   d) RNA only
9. SYBR Green is used to:
   a) Digest DNA
   b) Stain DNA ✅
   c) Break phosphodiester bonds
   d) Inhibit PCR
10. Supercoiled DNA migrates:
    a) Slower than linear DNA
    b) Faster than linear DNA ✅
    c) Same as linear DNA
    d) Cannot migrate

11. Higher agarose concentration:
    a) Better for small fragments ✅
    b) Better for large fragments
    c) No effect
    d) Degrades DNA
12. Voltage affects:
    a) Migration speed ✅
    b) DNA size
    c) DNA sequence
    d) Staining
13. TAE buffer is used to:
    a) Stain DNA
    b) Maintain pH and conductivity ✅
    c) Cut DNA
    d) Amplify DNA
14. PCR product verification uses:
    a) SDS-PAGE
    b) Agarose gel electrophoresis ✅
    c) Western blot
    d) Northern blot
15. RNA integrity is checked using:
    a) DNA gel
    b) RNA gel electrophoresis ✅
    c) Protein gel
    d) Lipid gel
16. Capillary electrophoresis is used for:
    a) Manual DNA separation
    b) Automated high-throughput DNA separation ✅
    c) Protein degradation
    d) Lipid analysis
17. DNA conformation affects migration because:
    a) Supercoiled DNA moves faster ✅
    b) Linear DNA moves faster
    c) No effect
    d) Only RNA is affected
18. Agarose concentration is typically:
    a) 0.1–0.5%
    b) 0.7–2% ✅
    c) 5–10%
    d) 10–15%
19. Visualization of DNA requires:
    a) Light
    b) UV light ✅
    c) Heat
    d) Electric field only

20. SDS-PAGE is used to separate:
    a) DNA
    b) RNA
    c) Proteins ✅
    d) Lipids
21. PFGE is important for:
    a) Small DNA only
    b) Genomic mapping of large DNA ✅
    c) RNA sequencing
    d) Protein folding
22. DNA ladder is used to:
    a) Visualize RNA
    b) Estimate fragment size ✅
    c) Stain proteins
    d) Digest DNA
23. Limitations of gel electrophoresis include:
    a) Inability to separate DNA
    b) Toxic dyes and time consumption ✅
    c) No visualization
    d) DNA degradation always
24. Gel electrophoresis is widely used in:
    a) Molecular biology ✅
    b) Lipid synthesis
    c) Photosynthesis
    d) Protein folding only

B. Short Questions (6 × 3 marks each)

1. Define gel electrophoresis and explain its principle.
2. Differentiate between agarose gel and polyacrylamide gel.
3. Describe the role of DNA dyes in electrophoresis.
4. Explain the effect of gel concentration on DNA separation.
5. List three applications of gel electrophoresis in molecular biology.
6. What is pulsed-field gel electrophoresis and when is it used?

C. Long Questions (6 × 5 marks each)

1. Explain the principle and procedure of agarose gel electrophoresis.
2. Discuss the types of gels used for DNA, RNA, and protein separation.
3. Describe factors affecting migration of DNA in gel electrophoresis.
4. Explain visualization methods of DNA in gels and their safety considerations.
5. Discuss applications of gel electrophoresis in molecular cloning and diagnostics.

6. Compare and contrast agarose gel electrophoresis, PAGE, and PFGE in terms of resolution, sample type, and application.

BIO5101 – Introduction to Biotechnology

Topic 24 – Southern, Northern & Western Blotting Techniques

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1. Introduction to Blotting Techniques

Blotting techniques are essential molecular biology methods used to detect specific nucleic acid or protein molecules from a complex mixture. These biochemical assays combine electrophoresis, transfer to membrane, and specific detection via probes or antibodies. They help researchers analyze gene expression, identify mutations, verify genetic modifications, and study proteins.

The three classical blotting techniques are:

Technique	Detects	Probe Type
Southern Blot	DNA	DNA or RNA probe
Northern Blot	RNA	DNA or RNA probe
Western Blot	Protein	Antibodies

Each method shares a common workflow:

- Separation of molecules by electrophoresis
- Transfer to a membrane
- Detection using complementary probes or specific antibodies

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2. Southern Blotting (DNA Blotting)

Purpose

1. Detecting specific DNA sequences in a mixture
2. Mutation analysis
3. Identification of restriction fragment length polymorphisms (RFLPs)
4. Verification of cloned genes
5. Forensic and paternity testing

Steps of Southern Blotting

- DNA Extraction
  Isolate genomic DNA from cells or tissues.
- Restriction Enzyme Digestion
  DNA is cut into fragments using restriction endonucleases.
- Agarose Gel Electrophoresis
  DNA fragments are separated based on size.
- Denaturation
  Gel is soaked in an alkaline solution to convert double-stranded DNA into single-stranded DNA.
- Transfer to Membrane
  DNA is transferred from gel to nitrocellulose or nylon membrane (capillary transfer, electrophoretic transfer).
- Hybridization
  A labeled DNA/RNA probe complementary to target sequence is added.
- Washing
  Removes unbound or weakly bound probes.
- Detection
  Radioactive, fluorescent, or enzymatic labels produce a visible signal.

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3. Northern Blotting (RNA Blotting)

Purpose

1. Detection of RNA molecules
2. Measurement of gene expression levels
3. Studying mRNA size and abundance
4. Evaluating transcription patterns

Steps

- RNA Extraction
- Denaturing Agarose Gel Electrophoresis
  Prevents secondary structures in RNA.
- Transfer to Membrane
- Hybridization with DNA/RNA probe
- Washing and Detection

Key Notes

- RNA is highly unstable → requires RNase-free environment.
- Indicates which genes are active in a cell.

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4. Western Blotting (Protein Blotting)

Purpose

- Detect specific proteins in a mixture
- Study protein expression, size, purity
- Identify post-translational modifications
- Used in medical diagnostics (e.g., HIV confirmation tests)

Steps

1. Protein Extraction
2. SDS-PAGE
   Proteins are separated by size using an SDS denaturing gel.
3. Transfer to PVDF/Nitrocellulose Membrane
4. Blocking
   Prevents nonspecific binding using milk or BSA.
5. Primary Antibody Binding
   Specific to the protein of interest.
6. Secondary Antibody Binding
   Enzyme-linked antibody that binds to primary antibody.
7. Detection
   Colorimetric, chemiluminescent, or fluorescent signals.

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5. Differences Between Southern, Northern & Western Blots

Feature	Southern	Northern	Western
Detects	DNA	RNA	Protein
Probe	DNA/RNA	DNA/RNA	Antibody
Gel Used	Agarose	Denaturing agarose	SDS-PAGE
Application	Gene detection	Gene expression	Protein expression

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6. Applications of Blotting Techniques

Southern Blot

1. Gene mapping

2. Detecting mutations
3. DNA fingerprinting
4. Verification of gene cloning

Northern Blot

1. Gene expression studies
2. mRNA size determination
3. Studying regulatory pathways

Western Blot

- Protein quantification
- Disease diagnostics
- Checking protein purification
- Studying protein modifications

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7. Advantages & Limitations

Southern

✔ Specific, reliable

✗ Time-consuming, requires radioactive probes

Northern

✔ Direct measurement of gene expression

✗ RNA is unstable, requires extreme care

Western

✔ Highly sensitive

✗ Antibody quality affects accuracy

---

8. Summary

Blotting techniques are foundational tools in biotechnology, enabling detection and analysis of DNA, RNA, and proteins. Southern, Northern, and Western blots each target different biomolecules, using specific probes and workflows for accurate detection.

A. 24 MCQs (1 mark each)

- Southern blot detects:
  a) RNA
  b) Protein

c) DNA ✅
d) Lipids

- Northern blot detects:
  a) DNA
  b) RNA ✅
  c) Protein
  d) Carbohydrates
- Western blot detects:
  a) Lipids
  b) DNA
  c) Protein ✅
  d) RNA
- Southern blot was developed by:
  a) Edwin Southern ✅
  b) Oliver Western
  c) Johnson Northern
  d) Watson
- SDS-PAGE is used in:
  a) Southern
  b) Northern
  c) Western ✅
  d) None
- DNA probes are used in:
  a) Western
  b) Southern and Northern ✅
  c) None
  d) All
- The membrane used in blotting is:
  a) Nylon
  b) Nitrocellulose
  c) PVDF
  d) All of the above ✅
- RNA requires a _____ environment:
  a) DNase-free
  b) RNase-free ✅
  c) Acidic
  d) Neutral

- Denaturation is essential in Southern blot to:
  a) Break proteins
  b) Make DNA single-stranded ✅
  c) Remove stains
  d) Increase size
- Hybridization involves:
  a) Protein folding
  b) Complementary base pairing of probe and target DNA/RNA ✅
  c) Oxidation
  d) Electrophoresis
- Western blot uses:
  a) DNA probe
  b) RNA probe
  c) Antibodies ✅
  d) Lipid dyes
- Northern blot is mainly used to study:
  a) Replication
  b) Transcription (gene expression) ✅
  c) Translation
  d) Metabolism
- Blocking is done in Western blot to prevent:
  a) Protein migration
  b) Non-specific antibody binding ✅
  c) RNA degradation
  d) DNA denaturation
- PVDF membrane is mainly used for:
  a) DNA
  b) RNA
  c) Protein detection (Western) ✅
  d) Lipids
- Southern blot includes:
  a) SDS-PAGE
  b) Agarose gel electrophoresis ✅
  c) None
  d) Both
- Radioactive probes are often used in:
  a) Northern

b) Southern
c) Both Northern and Southern ✅
d) Western

- Secondary antibodies bind to:
  a) DNA
  b) Primary antibody ✅
  c) RNA
  d) Probe
- High gene expression results in:
  a) Faint bands
  b) Strong RNA bands in Northern blot ✅
  c) No bands
  d) Dark DNA bands
- DNA fingerprinting uses:
  a) Western
  b) Northern
  c) Southern blotting ✅
  d) None
- RNase enzyme degrades:
  a) DNA
  b) RNA ✅
  c) Proteins
  d) Membranes
- mRNA is detected by:
  a) Southern
  b) Western
  c) Northern blotting ✅
  d) None
- Western blot confirms:
  a) HIV infection ✅
  b) Diabetes
  c) Malaria
  d) Cancer
- Denaturing agents used in Northern blot:
  a) SDS
  b) Urea/formaldehyde ✅

c) Ethanol
d) Acetone

- Target molecule of Western blot:
  a) DNA
  b) RNA
  c) Protein ✅
  d) Lipid

B. Short Questions (6 × 3 marks each)

- Define Southern blotting and mention its main steps.
- How does Northern blotting help in studying gene expression?
- Describe the role of probes in Southern and Northern blotting.
- Why is blocking needed in Western blotting?
- Explain the importance of SDS in SDS-PAGE.
- List three common applications of Western blotting in biotechnology.

C. Long Questions (6 × 5 marks each)

- Explain Southern blotting in detail, including its principle, procedure, and applications.
- Discuss the complete Northern blotting process and how it is used to study gene expression.
- Describe the Western blotting procedure, highlighting the role of primary and secondary antibodies.
- Compare Southern, Northern, and Western blotting in terms of method, probes, applications, and molecules detected.
- Explain probe labeling and hybridization mechanisms in nucleic acid blotting techniques.
- Provide detailed applications of blotting techniques in diagnostics, research, forensic science, and biotechnology.

BIO5101 – Introduction to Biotechnology

Topic 25 – DNA Sequencing Techniques

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1. Introduction to DNA Sequencing

DNA sequencing is the process of determining the **precise order of nucleotides** (A, T, G, C) within a DNA molecule.
It is fundamental to:

- Genome analysis
- Mutation detection
- Disease diagnostics
- Forensic analysis
- Evolutionary studies
- Biotechnology & genetic engineering

Modern sequencing has evolved from **first-generation sequencing (Sanger)** to **second-generation (Next-Generation Sequencing; NGS)** and **third-generation (single-molecule sequencing).**

---

2. First Generation Sequencing (Sanger Sequencing)

Principle

The method uses:

- **DNA polymerase**
- **Normal nucleotides (dNTPs)**
- **Chain-terminating nucleotides (ddNTPs)** lacking the 3' OH group

When a **ddNTP** is incorporated, DNA elongation stops.

Steps

1. **DNA denaturation**
2. **Primer binding to template**
3. **Chain elongation** with both dNTPs & ddNTPs
4. **Fragment separation** using gel electrophoresis or capillary electrophoresis
5. **Sequence analysis** by reading bands or fluorescent signals

Applications

- Single gene sequencing
- Mutation detection
- Small plasmid sequencing
- Clinical diagnostics

---

3. Maxam–Gilbert Sequencing (Chemical Sequencing)

Principle

- DNA is radioactively labeled
- Chemical treatments cleave DNA at specific bases
    - Dimethyl sulfate → G
    - Formic acid → A + G
    - Hydrazine → C + T

Disadvantages

- Hazardous chemicals
- Complex protocol
- Rarely used today

---

4. Second Generation Sequencing (Next Generation Sequencing – NGS)

NGS allows **massively parallel sequencing**, generating millions of reads at once.

Common NGS Platforms

- Illumina (Sequencing by synthesis)
- Ion Torrent (pH-based detection)
- Roche 454 (Pyrosequencing – discontinued)

General Workflow

1. **DNA fragmentation**
2. **Adaptor ligation**
3. **Library preparation**
4. **Amplification (bridge PCR/emulsion PCR)**
5. **Parallel sequencing**
6. **Bioinformatics analysis**

Advantages

- High-throughput
- Low cost per base
- Fast and efficient

Applications

- Whole Genome Sequencing
- Whole Exome Sequencing
- RNA Sequencing
- Metagenomics
- Transcriptomics

---

5. Third Generation Sequencing

Sequencing **single DNA molecules** without amplification.

Platforms

- PacBio SMRT sequencing
- Oxford Nanopore sequencing

Advantages

- Ultra-long reads
- Detects epigenetic modifications
- Real-time sequencing

---

6. Applications of DNA Sequencing

- Personalized medicine
- Cancer genomics
- Microbial identification
- Gene editing validation
- Evolutionary studies
- GMO testing

---

7. Limitations of Sequencing

- Cost (still high for large genomes)
- Data storage issues
- Bioinformatics complexity
- Errors in long-read technologies

Assessment Questions for TOPIC 25

24 MCQs (1 Mark Each) – With Answers

1. DNA sequencing determines the order of:
   A. Nucleotides ✔
2. Sanger sequencing uses:
   C. ddNTPs ✔
3. ddNTPs stop chain growth because they lack:
   B. 3' OH ✔
4. Maxam–Gilbert sequencing uses:
   A. Chemical cleavage ✔
5. NGS stands for:
   C. Next Generation Sequencing ✔
6. Illumina platform uses:
   D. Sequencing by synthesis ✔
7. Pyrosequencing detects release of:
   B. Pyrophosphate ✔
8. Ion Torrent detects changes in:
   A. pH ✔
9. Sanger sequencing is also called:
   C. Chain termination method ✔
10. NGS is characterized by:
    B. Massive parallel sequencing ✔
11. Nanopore sequencing detects changes in:
    A. Electrical current ✔
12. PacBio sequencing uses:
    C. SMRT cells ✔
13. Maxam–Gilbert requires:
    D. Radioactive labeling ✔
14. The fastest sequencing method is:
    B. Nanopore sequencing ✔
15. Sanger is suitable for:
    A. Single genes ✔
16. Illumina detects fluorescent:
    C. Signals ✔
17. DNA fragmentation is required for:
    D. NGS ✔

18. RNA sequencing is done using:
    B. NGS ✔
19. Longest read lengths are produced by:
    C. Nanopore ✔
20. Whole genome sequencing is mainly done by:
    B. NGS ✔
21. A limitation of Sanger sequencing:
    D. Low throughput ✔
22. Chain termination occurs due to:
    A. ddNTP incorporation ✔
23. Bioinformatics is essential for:
    C. NGS data analysis ✔
24. Chemical sequencing is also known as:
    B. Maxam–Gilbert method ✔

---

6 Short Questions (3 Marks Each)

1. What is the principle behind Sanger sequencing?
2. Compare Maxam–Gilbert and Sanger sequencing.
3. Describe the steps of NGS library preparation.
4. What are the advantages of third-generation sequencing?
5. Why is NGS considered high-throughput?
6. List three applications of DNA sequencing in biotechnology.

---

6 Long Questions (5 Marks Each)

1. Explain Sanger sequencing in detail with a labelled diagram.
2. Describe the complete workflow of Next Generation Sequencing (NGS).
3. Compare first, second, and third-generation sequencing technologies.
4. Discuss the applications of DNA sequencing in medicine and biotechnology.
5. Explain Illumina sequencing by synthesis with steps.
6. Write a detailed note on PacBio and Nanopore sequencing.

BIO5101 – Introduction to Biotechnology

Topic 26 – Polymerase Chain Reaction (PCR)

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1. Introduction to PCR

The Polymerase Chain Reaction (PCR) is a revolutionary molecular technique developed by Kary Mullis in 1983.
It allows rapid amplification of a specific DNA segment from a minimal starting sample.
PCR is essential in:

- Genetic engineering
- Diagnostics
- Forensic science
- Research
- Cloning
- Pathogen detection

With PCR, a single DNA molecule can be amplified into millions or billions of copies within a few hours.

---

2. Principle of PCR

PCR mimics the natural DNA replication process but occurs in vitro using:

6. Template DNA
7. Primers (short DNA sequences)
8. Thermostable DNA polymerase (Taq polymerase)
9. Deoxynucleotides (dNTPs)
10. Buffer solution
11. Thermal cycler

PCR follows a cycle of temperature changes that drive DNA denaturation, primer binding, and DNA synthesis.

---

3. Components of PCR

- Template DNA
  - The DNA region to be amplified.
- Primers
  - Short (18–25 nucleotides) sequences that flank the target DNA.
- dNTPs (A, T, G, C)
  - Raw materials for new DNA strand synthesis.
- Taq Polymerase
  - A thermostable enzyme isolated from Thermus aquaticus.
  - Can withstand high temperatures (95°C).
- Buffer & $MgCl_2$

- Maintain optimal pH and enzyme activity.

---

4. Steps of PCR

PCR consists of repetitive cycles. Each cycle doubles the DNA quantity.

1. Denaturation (94–95°C)

5. Double-stranded DNA melts into single strands.

2. Annealing (50–65°C)

- Primers bind to complementary sequences.

3. Extension/Elongation (72°C)

- Taq polymerase synthesizes new DNA strands.

One cycle = 3 steps

PCR usually runs 25–35 cycles, producing exponential amplification.

---

5. Types of PCR

1. Conventional PCR

- Basic amplification, analyzed by gel electrophoresis.

2. Real-Time PCR (qPCR)

8. Quantifies DNA in real time
9. Uses fluorescent dyes or probes

3. Reverse Transcriptase PCR (RT-PCR)

5. Converts RNA to cDNA
6. Used for RNA viruses (e.g., COVID-19 testing)

4. Multiplex PCR

4. Amplifies multiple DNA targets simultaneously using multiple primer pairs.

5. Nested PCR

- Increases specificity using two sets of primers.

6. Digital PCR

- Absolute quantification of DNA using microfluidics.

7. Touchdown PCR

- Uses gradually decreasing annealing temperatures for higher specificity.

---

6. Applications of PCR

- Disease diagnosis (viral/bacterial detection)

- Forensic DNA fingerprinting
- Gene cloning and genetic modification
- Paternity testing
- Detection of genetically modified organisms (GMOs)
- Environmental monitoring
- Mutation and SNP analysis
- Research in genomics and proteomics

---

7. Limitations of PCR

- Requires prior knowledge of target sequence
- Contamination can produce false positives
- Primer design is critical
- Taq polymerase may cause errors (no proofreading)
- Inhibitors in samples can affect amplification

— END OF TOPIC CONTENT —

---

24 MCQs (1 Mark Each) – With Answers

1. PCR stands for:
   C. Polymerase Chain Reaction ✔
2. PCR was invented by:
   A. Kary Mullis ✔
3. PCR amplifies:
   B. DNA ✔
4. Thermostable polymerase used in PCR:
   C. Taq polymerase ✔
5. Taq polymerase is extracted from:
   D. Thermus aquaticus ✔
6. Denaturation temperature is around:
   A. 94–95°C ✔
7. Annealing allows:
   B. Primers to bind ✔
8. Extension temperature is:
   D. 72°C ✔

9. Number of PCR cycles usually:
   C. 25–35 ✔
10. RT-PCR starts with:
   A. RNA ✔
11. qPCR measures:
   C. DNA quantity in real time ✔
12. Multiplex PCR uses:
   B. Multiple primer pairs ✔
13. Nested PCR improves:
   A. Specificity ✔
14. PCR product is analyzed by:
   D. Gel electrophoresis ✔
15. dNTPs are required for:
   C. DNA synthesis ✔
16. Touchdown PCR improves:
   B. Specificity ✔
17. Digital PCR is used for:
   A. Absolute quantification ✔
18. PCR contamination causes:
   D. False positives ✔
19. Real-Time PCR detects:
   C. Fluorescence ✔
20. Key requirement for PCR is:
   B. Primers ✔
21. PCR exponential amplification means product doubles:
   A. Every cycle ✔
22. Limitation of Taq polymerase:
   D. No proofreading ability ✔
23. First step in PCR cycle:
   A. Denaturation ✔
24. PCR cannot amplify without:
   B. Template DNA ✔

---

6 Short Questions (3 Marks Each)

- Describe the three basic steps of PCR.

- What is the role of primers in PCR?
- Compare conventional PCR and real-time PCR.
- Why is Taq polymerase essential for PCR?
- Explain the principle of reverse transcriptase PCR.
- List three major applications of PCR in biotechnology.

---

6 Long Questions (5 Marks Each)

- Explain the complete mechanism of PCR with a diagram.
- Discuss all major types of PCR with examples and applications.
- Describe real-time PCR (qPCR) in detail and explain its importance.
- Explain the components of a PCR reaction mixture and their functions.
- Discuss the limitations of PCR and how they can be minimized.
- Explain the importance of PCR in disease diagnosis, especially viral detection.

Topic 27: Quaternary Structure of Proteins

(BIO5101 – Introduction to Biotechnology)

Detailed Explanation

Proteins are essential biological macromolecules made up of amino acids. Most proteins fold into complex structures to become biologically functional. Protein structure is described in four hierarchical levels:

1. **Primary Structure** – sequence of amino acids
2. **Secondary Structure** – α-helices, β-sheets
3. **Tertiary Structure** – overall 3D shape of a single polypeptide
4. **Quaternary Structure** – association of **two or more polypeptide chains**

What Is Quaternary Structure?

The **Quaternary Structure** refers to the highest level of protein organization where **multiple polypeptide chains**, called **subunits**, interact to form a functional protein complex.

These subunits may be:

- **Identical** → *homomers* (e.g., lactate dehydrogenase)
- **Different** → *heteromers* (e.g., hemoglobin)

Why do proteins form quaternary structures?

They allow:

- Increased **stability**
- **Regulation** of activity
- Formation of **multi-functional complexes**
- Increased **efficiency** (cooperativity)

Characteristics of Quaternary Structure

1. Subunit Composition

Proteins can have 2, 3, 4, or many subunits:

- **Dimer** → 2 subunits
- **Trimer** → 3 subunits
- **Tetramer** → 4 subunits
- **Multimer** → many subunits

Example:

- **Hemoglobin** → tetramer (2α + 2β subunits)

---

2. Types of Bonds Stabilizing Quaternary Structure

Subunits interact via:

Non-covalent interactions

- Hydrogen bonds
- Hydrophobic interactions
- Ionic (electrostatic) interactions
- Van der Waals forces

Covalent Interaction

- **Disulfide bonds** (S–S linkages) between cysteine residues

---

3. Cooperativity

Some quaternary proteins show **allosteric behavior**:

- Binding of a ligand to one subunit **affects activity of other subunits**
- Example: **Hemoglobin oxygen-binding cooperativity**

---

4. Functional Significance

Quaternary proteins allow:

- **Allosteric regulation**
- Increased **binding efficiency**
- Division of labor between subunits
- Structural support
- Complex metabolic regulation

Examples:

- **DNA polymerase** (multi-subunit enzyme)
- **ATP synthase** (rotary motor protein)
- **Immunoglobulins** (antibody tetramers)

Examples of Proteins with Quaternary Structure

Protein	Subunits	Significance
Hemoglobin	4 (2α + 2β)	Oxygen transport
Immunoglobulins (IgG)	4 (2 heavy + 2 light chains)	Immune defense
ATP Synthase	Multisubunit	ATP synthesis
Collagen	3 polypeptide chains	Structural protein
RNA Polymerase	Multi-subunit enzyme	Transcription

Importance in Biotechnology

- Quaternary proteins are targets for **drug design**
- Monoclonal antibodies rely on quaternary interactions
- Industrial enzymes are engineered by altering subunit interfaces
- Recombinant proteins often require correct subunit assembly
- Mutations in subunit interfaces cause diseases (e.g., sickle cell anemia)

Disorders Related to Quaternary Structure

1. **Sickle Cell Anemia**
   - Caused by mutation in β-subunit of hemoglobin
   - Distortion of quaternary structure → sickle-shaped red blood cells
2. **Alzheimer's Disease**
   - Improper assembly of Amyloid-β proteins → aggregates
3. **Prion Diseases**
   - Misfolded prion proteins affect other subunits → neurodegeneration

---

Summary

- Quaternary structure = assembly of multiple polypeptide subunits
- Stabilized by covalent & non-covalent interactions
- Important for cooperativity, regulation, stability
- Common in enzymes, antibodies, structural proteins
- Essential for biotechnology applications

---

24 MCQs (with Answers)

1. Quaternary structure refers to:

a) Arrangement of amino acids
b) Folding of a single polypeptide
c) Interaction of multiple polypeptide chains
d) Hydrogen bonding only
Answer: c

2. Subunits in quaternary structure are called:

a) Domains
b) Tetramers
c) Polypeptide chains
d) Motifs
Answer: c

3. Hemoglobin contains how many subunits?

a) 2
b) 3
c) 4
d) 6
Answer: c

4. A protein with identical subunits is called:

a) Heteromer
b) Homomer
c) Trimer
d) Monomer
Answer: b

5. Which bond is covalent in quaternary structure?

a) Hydrogen bond
b) Ionic bond
c) Disulfide bond
d) Van der Waals
Answer: c

6. The smallest multimer with 3 subunits is called:

a) Dimer
b) Trimer
c) Tetramer
d) Pentamer
Answer: b

7. Hemoglobin is an example of:

a) Monomer
b) Dimer
c) Tetramer

d) Polymer
Answer: c

8. Immunoglobulin has how many chains?

a) 2
b) 3
c) 4
d) 5
Answer: c

9. Allosteric regulation is common in:

a) Primary structure
b) Secondary structure
c) Quaternary structure
d) None
Answer: c

10. Sickle cell anemia affects:

a) DNA
b) Hemoglobin
c) Collagen
d) ATP synthase
Answer: b

11. Subunit interaction in quaternary structure is mostly:

a) Non-covalent
b) Covalent
c) Metallic
d) Radiation
Answer: a

12. Example of structural quaternary protein:

a) Hemoglobin
b) Collagen
c) Myosin
d) Keratin
Answer: b

13. Two or more different subunits =:

a) Homomer
b) Heteromer
Answer: b

14. Cooperative binding occurs in:

a) Insulin
b) Hemoglobin
c) Trypsin
d) Keratin
Answer: b

15. Quaternary structure is lost during:

a) Hydrolysis
b) Denaturation
c) Mutation
d) Polymerization
Answer: b

16. Disulfide bonds form between:

a) Lysine
b) Arginine
c) Cysteine
d) Glutamine
Answer: c

17. Collagen contains how many chains?

a) 1
b) 2
c) 3
d) 4
Answer: c

18. The enzyme ATP synthase is:

a) Single protein
b) Multi-subunit complex
c) Lipid
d) RNA
Answer: b

19. Quaternary structure increases:

a) Solubility
b) Efficiency
c) Cooperativity
d) All of these
Answer: d

20. Subunit dissociation affects:

a) Function
b) Stability
c) Regulation
d) All of these
Answer: d

21. IgG has how many heavy chains?

a) 1
b) 2
c) 3
d) 4
Answer: b

22. Misfolding of quaternary proteins leads to:

a) Cancer
b) Prion diseases
c) Diabetes
d) Cataract
Answer: b

23. A protein with 4 different subunits is a:

a) Homotetramer
b) Heterotetramer
c) Dimer
d) Monomer
Answer: b

24. Which force is NOT involved in quaternary structure?

a) Hydrogen bonding
b) Ionic interaction
c) Disulfide bond

d) Peptide bond formation between chains
Answer: d

---

6 Short Questions (3 Marks Each)

1. Define quaternary structure and give one example.
2. Differentiate between homomeric and heteromeric proteins.
3. List three non-covalent interactions stabilizing quaternary structure.
4. Briefly explain cooperative binding with the example of hemoglobin.
5. How do disulfide bonds contribute to quaternary stability?
6. Mention three biological proteins that possess quaternary structure.

---

6 Long Questions (5 Marks Each)

1. Describe in detail the quaternary structure of proteins with suitable examples.
2. Explain the forces and bonds involved in stabilizing quaternary structures.
3. Discuss the quaternary structure of hemoglobin and its cooperativity mechanism.
4. Elaborate the functional advantages of quaternary structure in enzymes and antibodies.
5. Explain with examples how disorders arise due to defective quaternary structure.
6. Discuss the significance of quaternary structure in biotechnology and protein engineering.

Topic 28: Genetic Materials in Viruses – Detailed Explanation

Viruses are **acellular infectious particles** that depend entirely on host cells for replication. Unlike living organisms, viruses do not contain both DNA and RNA simultaneously. Their **genetic material (genome) is either DNA or RNA**, never both. Viral genomes show remarkable diversity in type, structure, organization, and replication strategies.

---

1. Nature of Viral Genetic Material

Viruses exhibit **all possible variations** in genetic material:

A. Based on Type of Nucleic Acid

1. **DNA Viruses**

- Contain DNA as genetic material.
- DNA can be:
  - **Double-stranded DNA (dsDNA)** → e.g., Adenovirus, Herpesvirus, Poxvirus
  - **Single-stranded DNA (ssDNA)** → e.g., Parvovirus
- Generally replicate inside the **nucleus** of host cells (except poxviruses).

2. **RNA Viruses**
   - Contain RNA as the genetic material.
   - RNA can be:
     - **Double-stranded RNA (dsRNA)** → e.g., Rotavirus
     - **Single-stranded RNA (ssRNA)** (most common)
     - ssRNA may be:
       - **Positive-sense (+) RNA** → can directly act as mRNA (e.g., Poliovirus)
       - **Negative-sense (–) RNA** → complementary to mRNA; needs viral RNA polymerase (e.g., Influenza virus)
       - **Ambisense RNA** → genome has both + and – regions (e.g., Arenaviruses)

---

2. Genome Size in Viruses

Viral genomes vary significantly in size:

- **Small genome:** Parvovirus (~5 kb)
- **Medium genome:** Adenovirus (~36 kb)
- **Large genome:** Poxvirus (~200–300 kb)
- **Giant viruses:** Mimivirus (~1,200 kb), Megavirus (~1,259 kb)

Smaller genomes encode fewer proteins and depend more heavily on host machinery.

---

3. Configuration of Viral Genetic Material

A. Linear vs Circular Genomes

- **Linear DNA/RNA:** e.g., Adenovirus, Herpesvirus, Paramyxovirus
- **Circular DNA:** e.g., Papillomavirus, Polyomavirus
- **Circular RNA:** uncommon in typical animal viruses but exists in viroids (plants)

B. Segmented vs Non-Segmented Genomes

- **Segmented genomes:** Genome is divided into multiple fragments.
  - Example: Influenza virus (8 segments), Rotavirus (11 segments)

    - Advantage: Allows **genetic reassortment**, generating new strains.
- **Non-segmented genomes:** Single continuous molecule
    - Example: Measles virus, Rabies virus, Coronaviruses

---

4. Haploid vs Diploid Viruses

Most viruses carry a **single copy** of their genome (haploid).
Exception:

- **Retroviruses** (e.g., HIV) contain **two identical copies of RNA** → technically **diploid viruses**.

---

5. Packaging of Viral Genetic Material

Viral genomes are tightly packed due to their small capsid volume:

● **DNA packaged with viral or host-derived proteins** → stabilizes genome
● **RNA genomes often have secondary structures** (hairpins, loops) that compact them
● Viral nucleic acids are surrounded by:

- **Capsid** (protein coat)
- Sometimes **envelope** derived from host membrane

---

6. Genetic Variability in Viruses

Viruses evolve rapidly due to:

A. High Mutation Rates

- Especially RNA viruses (no proofreading by RNA polymerase)

B. Reassortment

- Occurs in segmented viruses during co-infection (e.g., Influenza pandemic strains)

C. Recombination

- Exchange of genetic material between viral genomes

- Common in retroviruses and DNA viruses

These variations lead to:

- Emergence of new viral strains
- Drug resistance
- Host range changes

---

7. Special Case: Retroviral Genetic Material

Retroviruses (HIV, HTLV):

- Contain **two copies of (+) ssRNA**
- Use **reverse transcriptase** to convert RNA → DNA
- Integrate into host genome (forming **provirus**)
- Have regulatory genes (gag, pol, env)

Retroviruses use the **RNA → DNA → RNA** pathway, unlike typical viruses.

---

8. Summary

Viral genetic material is:

- Diverse in type (DNA or RNA)
- Variable in structure (ss, ds; linear, circular)
- Packaged tightly in capsids
- Highly mutable
- Adapted for rapid replication and evolution

This flexibility makes viruses successful pathogens.

---

24 MCQs (1 mark each)

(Correct answers included)

#	Question	Options	Answer
1	Viruses contain which type of genetic material?	a) DNA only b) RNA only c) Either DNA or RNA d) Both DNA & RNA	c
2	Which virus contains ssDNA?	a) Parvovirus b) Herpesvirus c) Poxvirus d) HIV	a
3	Positive-sense RNA acts directly as:	a) rRNA b) tRNA c) mRNA d) DNA template	c
4	Which virus has dsRNA?	a) Rabies b) Rotavirus c) HIV d) Influenza	b
5	Retroviruses contain how many RNA copies?	a) One b) Two c) Three d) Four	b
6	Largest viral genome is found in:	a) Parvovirus b) Mimivirus c) Rabies virus d) Adenovirus	b
7	Negative-sense RNA requires:	a) Reverse transcriptase b) DNA polymerase c) RNA-dependent RNA polymerase d) Ligase	c
8	Influenza virus genome is:	a) Circular b) Linear c) Segmented d) Both b & c	d
9	Genetic reassortment occurs in:	a) Non-segmented viruses b) Segmented viruses c) ssDNA viruses d) Retroviruses	b
10	Plant-like circular RNAs are found in:	a) Bacteriophages b) Viroids c) Retroviruses d) Poxviruses	b
11	The capsid encloses:	a) Lipid bilayer b) Genome c) Ribosomes d) Mitochondria	b
12	Which virus uses reverse transcriptase?	a) Adenovirus b) HIV c) Rabies d) Influenza	b
13	DNA viruses usually replicate in the:	a) Cytoplasm b) Golgi body c) Nucleus d) ER	c
14	Poxviruses replicate in the:	a) Cytoplasm b) Nucleus c) Mitochondria d) Ribosomes	a
15	Which is the smallest viral genome?	a) Adenovirus b) Parvovirus c) Poxvirus d) Mimivirus	b
16	Which genome type is rare in viruses?	a) ssRNA b) dsRNA c) ssDNA d) Circular RNA	d
17	Hairpin loops in RNA help in:	a) Translation b) Genome compaction c) Transcription d) Mutation	b

#	Question	Options	Answer
18	Retroviral DNA integrated in host genome is called:	a) Prophage b) Provirus c) Episome d) Plasmid	b
19	A virus with single continuous RNA molecule is:	a) Rotavirus b) Influenza c) Coronavirus d) HIV	c
20	Mutation rate is highest in:	a) DNA viruses b) RNA viruses c) Viroids d) Bacteriophages	b
21	Genome size of adenovirus is approx.:	a) 5kb b) 36kb c) 200kb d) 1200kb	b
22	Reverse transcription converts:	a) DNA→RNA b) RNA→DNA c) Protein→DNA d) RNA→Protein	b
23	Recombination in viruses leads to:	a) More ribosomes b) Increased stability c) Genetic diversity d) Loss of genome	c
24	Giant viruses like Mimivirus fall under:	a) RNA viruses b) Miniviruses c) Nucleocytoplasmic large DNA viruses d) Retroviruses	c

---

6 Short Questions (3 marks each)

1. Differentiate between positive-sense and negative-sense RNA viruses.
2. What is genome segmentation in viruses? Give two examples.
3. Explain why retroviruses are considered diploid.
4. Describe the role of hairpin structures in viral RNA packaging.
5. Why do RNA viruses mutate faster than DNA viruses?
6. Briefly explain genetic reassortment and its significance.

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6 Long Questions (5 marks each)

1. Describe the diversity of viral genetic material with examples of each type.
2. Compare dsDNA viruses, ssDNA viruses, dsRNA viruses, and ssRNA viruses in terms of structure, replication, and examples.
3. Explain in detail how retroviruses replicate, focusing on reverse transcription and integration.
4. Discuss genome segmentation in viruses and how it contributes to viral evolution.
5. Describe the factors responsible for high genetic variability in viruses (mutation, recombination, reassortment).

6. Explain the different configurations of viral genomes (linear, circular, segmented) with examples of each and their biological implications.

Topic 29: Organization of Genetic Material in Bacteria – 1

Bacteria are **prokaryotic organisms** and possess a simple but highly efficient genetic system. Their genetic material is compact, well-organized, and optimized for fast growth and rapid adaptation. Understanding bacterial genome organization provides the foundation for molecular biology, biotechnology, recombinant DNA technology, and microbial genetics.

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1. General Features of Bacterial Genetic Material

Bacteria typically contain:

1. **A single, circular chromosome (nucleoid)**
2. **Plasmids** (extrachromosomal DNA)
3. Occasionally **bacteriophages** integrated into DNA (prophages)

Bacterial DNA is **not enclosed by a nuclear membrane**. Instead, it exists in a specialized region known as the **nucleoid**.

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2. Nucleoid – The Bacterial Chromosome Organization

A. Shape and Structure

- Usually **circular**, **double-stranded DNA (dsDNA)**
- Some bacteria have **linear chromosomes** (rare, e.g., *Borrelia burgdorferi*)

B. Chromosome Size

- Varies from **0.6 Mbp to 10 Mbp**
  - *Mycoplasma genitalium*: ~0.6 Mb
  - *E. coli*: ~4.6 Mb
  - *Streptomyces*: up to 10 Mb

C. Supercoiling

Bacterial DNA is tightly packed by:

- **Supercoiling** (positive or negative)
- Enzymes:
  - **DNA gyrase**
  - **Topoisomerase I**

This helps fit long DNA into the small cell.

D. Nucleoid-Associated Proteins (NAPs)

Bacteria lack histones but use proteins like:

- HU
- IHF
- Fis
- H-NS

These proteins:

- Compact DNA
- Regulate DNA replication
- Organize DNA loops
- Assist gene expression

The nucleoid is dynamic and continuously rearranges during replication and cell division.

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3. Packaging of DNA in Bacteria

A. Loop Domains

Bacterial chromosomes are divided into **~40–100 loops (domains)**.
Each loop is independently supercoiled.

B. DNA Supercoiling

- Maintained by:
  - **DNA gyrase (adds negative supercoils)**
  - **Topoisomerase I (removes negative supercoils)**

Negative supercoiling:

- Aids replication
- Facilitates transcription
- Helps strand separation

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4. Gene Organization in Bacteria

A. Coding Density

Bacteria have **high gene density**:

- 85–90% of genome codes for proteins
- Very few introns

B. Operons

Bacterial genes are often arranged in **operons**:

- Group of genes transcribed together from a **single promoter**
- Produce **polycistronic mRNA**

Example:

- **lac operon**
- **trp operon**

Operons allow bacteria to regulate related genes simultaneously.

C. No Introns (mostly)

Prokaryotic genes generally lack introns, so transcription and translation occur rapidly.

---

5. Additional Genetic Elements in Bacteria

A. Plasmids

Small, circular, extrachromosomal DNA molecules:

- Self-replicating
- Carry genes for:
    - Antibiotic resistance
    - Toxin production
    - Fertility (F plasmid)

B. Transposons

DNA segments that can move within the genome.

C. Integrons

Genetic elements that capture and incorporate gene cassettes, often carrying antibiotic resistance genes.

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6. Bacterial Chromosomal Replication Features

- **Single Origin of Replication (OriC)**
- Bidirectional replication
- Very fast replication (e.g., *E. coli* doubles in 20 minutes)
- Replication coupled with transcription and translation

The simplicity of bacterial genome organization contributes to:

- Fast adaptation
- Efficient regulation
- Rapid growth
- Suitability for biotech applications

Summary

Bacterial genetic material is organized into:

- A circular, supercoiled chromosome
- Compact nucleoid structure with NAPs
- Additional elements like plasmids, transposons, and integrons
- Genes arranged in operons for efficient regulation

This structure supports bacterial survival and biotechnology applications.

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24 MCQs (1 mark each)

(Correct answers included)

#	Question	Options	Answer
1	Bacterial genetic material is located in the:	a) Nucleus b) Nucleoid c) Mitochondria d) Ribosomes	b

#	Question	Options	Answer
2	Most bacteria have which type of chromosome?	a) Linear dsDNA b) Circular dsDNA c) Circular ssDNA d) Linear dsRNA	b
3	DNA compaction in bacteria mainly depends on:	a) Histones b) NAPs c) Tubulin d) Actin	b
4	Which enzyme introduces negative supercoils?	a) Helicase b) DNA gyrase c) Ligase d) Primase	b
5	Operons produce:	a) Monocistronic RNA b) Polycistronic RNA c) Introns d) mRNA + tRNA mix	b
6	A typical genome size of *E. coli* is:	a) 0.6 Mb b) 4.6 Mb c) 10 Mb d) 12 Mb	b
7	The main packaging proteins in bacteria are:	a) Histones b) NAPs c) Nucleosomes d) Centrosomes	b
8	DNA loops in bacteria are:	a) 1–5 b) 10–20 c) 40–100 d) 200–500	c
9	Plasmids are:	a) RNA genomes b) Extrachromosomal DNA c) Chromosomes d) Ribosomes	b
10	A group of genes with a single promoter is called:	a) Operon b) Cistron c) Exon d) Regulon	a
11	The origin of replication in bacteria is:	a) TerC b) OriC c) OriT d) OriR	b
12	Which bacteria have linear chromosomes?	a) Salmonella b) Borrelia c) Bacillus d) Pseudomonas	b
13	NAP that introduces bends in DNA is:	a) Fis b) HU c) IHF d) H-NS	c
14	Replication in bacteria is:	a) Unidirectional b) Bidirectional c) Random d) In cytoplasm only	b
15	High coding density in bacteria means:	a) Many introns b) Few introns c) 50% genome coded d) No RNA genes	b
16	Plasmids replicate:	a) Dependently b) Independently c) Only with chromosome d) Only under stress	b
17	Which structure does *not* exist in bacteria?	a) Nucleoid b) Plasmid c) Histone-wrapped nucleosome d) Operon	c
18	Example of an antibiotic resistance element:	a) F plasmid b) R plasmid c) Ti plasmid d) Col plasmid	b

#	Question	Options	Answer
19	Transposons are also known as:	a) Jumping genes b) Sticky genes c) Silent genes d) Housekeeping genes	a
20	Which factor helps in compacting the nucleoid?	a) Ribosome b) DNA supercoiling c) Cell membrane d) Lysosomes	b
21	In bacteria, transcription and translation are:	a) Separated b) Coupled c) Absent d) Post-transcriptional	b
22	The bacterial genome lacks:	a) Exons b) Introns c) Promoters d) Coding sequences	b
23	Gene-rich regions in bacteria are due to:	a) Short introns b) No introns c) Long UTRs d) Multiple enhancers	b
24	Which plasmid is associated with bacterial conjugation?	a) R plasmid b) F plasmid c) Col plasmid d) Ti plasmid	b

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6 Short Questions (3 marks each)

1. Define nucleoid and explain its main structural properties.
2. What are Nucleoid Associated Proteins (NAPs)? Give examples and functions.
3. Explain operons with a suitable example.
4. Describe the significance of DNA supercoiling in bacteria.
5. What are plasmids? Write their two important roles.
6. Compare circular and linear bacterial chromosomes.

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6 Long Questions (5 marks each)

1. Describe the structure and organization of the bacterial nucleoid in detail.
2. Explain operon organization in bacteria and discuss its advantages with examples.
3. Discuss supercoiling in bacterial DNA, the enzymes involved, and its biological importance.
4. Differentiate between chromosomal DNA and plasmid DNA in bacteria with examples.
5. Explain the various genetic elements present in bacterial cells (chromosome, plasmids, transposons, integrons).
6. Describe how bacterial genome organization supports rapid growth and adaptability..

Topic 30: Organization of Genetic Material in Bacteria – 2

Bacterial genetic material is compact, efficient, and organized to allow rapid replication and adaptation. This part focuses on **plasmids, transposable elements, phages, and other mobile genetic elements** that add diversity and functional capabilities to bacteria beyond the main chromosome.

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1. Plasmids

Plasmids are **small, circular, extrachromosomal DNA molecules** that replicate independently of the bacterial chromosome. They are not essential for survival under normal conditions but provide **selective advantages** in specific environments.

Types of Plasmids

1. **Fertility (F) Plasmids**
   - Carry genes for **conjugation** (transfer between bacteria)
   - Contain **tra genes** for pilus formation
2. **Resistance (R) Plasmids**
   - Encode resistance to **antibiotics** or heavy metals
   - Can be transferred via conjugation
3. **Col Plasmids**
   - Encode **bacteriocins**, proteins that kill other bacteria
4. **Virulence Plasmids**
   - Contain genes enhancing **pathogenicity**, e.g., toxins
5. **Metabolic Plasmids**
   - Carry genes for **catabolism of unusual compounds**

Features of Plasmids

- Usually **circular**, double-stranded DNA
- Size ranges from **1 kb to 200 kb**
- Can be **conjugative** (self-transmissible) or **non-conjugative**

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2. Transposable Elements (Jumping Genes)

Transposable elements can **move within or between DNA molecules**. They play a key role in genome plasticity.

Types

1. **Insertion Sequences (IS Elements)**
    - Simplest transposons
    - Encode only **transposase** enzyme
    - Flanked by **inverted repeats (IRs)**
2. **Composite Transposons**
    - Carry **resistance genes** between IS elements
    - Can move as a single unit
3. **Non-Composite Transposons**
    - Carry functional genes without flanking IS elements
    - Transpose using their own transposase

Importance

- Contribute to **antibiotic resistance**
- Enable **genetic rearrangements**
- Facilitate horizontal gene transfer

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3. Integrons

Integrons are **genetic platforms** that can **capture and express gene cassettes**, often carrying **antibiotic resistance genes**.

- Contain **integrase enzyme** for recombination
- Have **promoter sequences** to express integrated genes
- Commonly found on plasmids and transposons

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4. Bacteriophages and Prophages

Bacteriophages are **viruses that infect bacteria**. Some integrate into the bacterial genome as **prophages**, contributing new traits.

- **Lysogenic cycle**: phage DNA integrates into host chromosome
- **Lytic cycle**: phage replicates independently, lyses the cell
- Prophages can encode **virulence factors**, e.g., toxins

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5. Genomic Islands

- Large DNA segments acquired via **horizontal gene transfer**
- Often **contain genes for pathogenicity, metabolism, or resistance**
- Flanked by **direct repeats** or **mobile elements**

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6. Plasmid-Host Interaction

- **Plasmid copy number**: Number of plasmids per cell varies (1–100)
- **Compatibility groups**: Plasmids with similar replication systems cannot coexist
- Plasmids influence bacterial **adaptation, virulence, and survival**

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7. Summary

Bacterial genomes consist of:

- **Chromosomal DNA**: single circular nucleoid
- **Plasmids**: independent DNA elements
- **Transposons**: mobile DNA for genetic rearrangement
- **Integrons**: gene capture and expression platforms
- **Prophages**: viral DNA contributing traits

This organization allows bacteria to **adapt quickly, acquire new traits, and survive environmental pressures**.

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24 MCQs (1 mark each)

#	Question	Options	Answer
1	Plasmids are usually:	a) Linear DNA b) Circular DNA c) RNA d) Protein	b
2	Which plasmid carries conjugation genes?	a) R plasmid b) F plasmid c) Col plasmid d) Virulence plasmid	b
3	R plasmids provide:	a) Metabolic genes b) Antibiotic resistance c) Virulence d) Fertility	b
4	Composite transposons are flanked by:	a) Promoters b) Inverted repeats c) Ribosomes d) tRNA	b

#	Question	Options	Answer
5	Integrons contain:	a) RNA polymerase b) Integrase enzyme c) DNA gyrase d) Helicase	b
6	Bacteriophages infect:	a) Eukaryotic cells b) Bacteria c) Viruses d) Fungi	b
7	Prophage DNA is integrated during:	a) Lytic cycle b) Lysogenic cycle c) Plasmid replication d) Transformation	b
8	Plasmids with similar replication systems cannot coexist due to:	a) Antibiotic resistance b) Compatibility c) Mutation d) Conjugation	b
9	Insertion sequences encode:	a) Integrase b) Transposase c) Ligase d) Polymerase	b
10	Metabolic plasmids provide:	a) Antibiotic resistance b) Fertility c) Catabolism of unusual compounds d) Virulence	c
11	Which plasmid carries toxin genes?	a) F plasmid b) Virulence plasmid c) R plasmid d) Col plasmid	b
12	Plasmid copy number can vary between:	a) 1–10 b) 1–100 c) 100–500 d) 500–1000	b
13	Integrons are often found on:	a) Ribosomes b) Plasmids and transposons c) Mitochondria d) Cytoplasm	b
14	Lysogenic phages can confer:	a) Virulence factors b) Replication speed c) Ribosomal RNA d) Transposons	a
15	Non-composite transposons carry:	a) Only transposase b) Functional genes c) Promoters only d) Ribosomal RNA	b
16	Col plasmids encode:	a) Conjugation proteins b) Bacteriocins c) Resistance d) Metabolic enzymes	b
17	Horizontal gene transfer contributes to:	a) Antibiotic resistance b) DNA replication c) Cell division d) Protein synthesis	a
18	Bacterial chromosomes are mostly:	a) Linear b) Circular c) RNA d) Segmented	b
19	Which is a mobile genetic element in bacteria?	a) Nucleoid b) Transposon c) Ribosome d) Cytoplasm	b
20	F plasmids are:	a) Non-conjugative b) Conjugative c) Antibiotic-resistant d) Metabolic	b

#	Question	Options	Answer
21	Genomic islands often carry:	a) Housekeeping genes b) Pathogenicity genes c) Ribosomes d) tRNAs	b
22	Prophages can integrate via:	a) Random insertion b) Site-specific recombination c) Ligase action d) Translation	b
23	Transposons help in:	a) Protein synthesis b) Genetic rearrangement c) Membrane synthesis d) Ribosome assembly	b
24	Virulence plasmids are important for:	a) Antibiotic resistance b) Pathogenicity c) DNA replication d) Transcription	b

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6 Short Questions (3 marks each)

1. Define plasmids and list their types.
2. What are transposable elements and why are they important?
3. Explain the role of integrons in bacterial genomes.
4. Describe lysogenic and lytic cycles of bacteriophages.
5. What are genomic islands and what function do they serve?
6. Explain plasmid copy number and compatibility groups.

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6 Long Questions (5 marks each)

1. Discuss the structure, types, and functions of plasmids in bacteria.
2. Explain transposable elements, their types, and their role in bacterial genetic variability.
3. Describe integrons and their role in antibiotic resistance.
4. Explain the lysogenic and lytic cycles of bacteriophages with examples.
5. Discuss genomic islands and their contribution to bacterial evolution.
6. Explain how mobile genetic elements contribute to horizontal gene transfer and bacterial adaptability